

SIEMENS S7-1500 · SCL FUNCTION BLOCK

FB_PPC_Controller

Plant Power Controller — I/O reference for commissioning and simulation testing. Covers all VAR_INPUT and VAR_OUTPUT, key DB39 parameters that govern behavior, and recommended test sequences.

VERSION	INSTANCE DB	CALLED FROM	PLANT
0.1	"PPC_Controller"	OB30 (cyclic, 100 ms)	CEF Tandarei · 48 MW · 10 × SMA SC

OB30 CALL PREREQUISITES

- 1. Call `FC_SkidElectricStatus` once, passing `"PPC_Controller".Skids[0..9]` — reads global DI tags and sets `ELECTRIC_OK` for each skid.
- 2. Call `FB_PPC_Controller` with all inputs wired. In **Sim_Mode** the switchgear and inverter hardware checks are bypassed automatically — no manual forcing required.
- 3. The IEC watchdog input (`IEC_Watchdog`) must toggle at least once within `Watchdog_Timeout` (default 30 s). In simulation, wire a clock bit or drive it from SCADA to prevent spurious FALLBACK.

VAR_INPUT 13 parameters			
PARAMETER	TYPE	DESCRIPTION	VALID RANGE / VALUES
EN_PPC	BOOL	Master enable. FALSE immediately commands all inverters to Stop (OperMode=303) and resets all ramps. Must be TRUE before any control output is produced.	FALSE / TRUE
IEC_Watchdog	BOOL	Heartbeat bit from SCADA/IEC comms thread. Must produce a rising edge at least once per <i>Watchdog_Timeout</i> (30 s default). If it stops toggling, Plant_Mode → FALLBACK and inverters stop.	Toggling BOOL Min 1 edge / 30 s
CycleTime_s	REAL	OB30 cycle time in seconds. Used by RampControl and QCapability to	0.05 ... 1.0 s Typical: 0.1

PARAMETER	TYPE	DESCRIPTION	VALID RANGE / VALUES
		compute per-scan step limits. Must match the actual OB30 interval.	
Cmd_P	REAL	Active power setpoint from SCADA or HMI (LOCAL mode), in kW. Passed through RampControl (rate-limited), then to FreqResponse (droop added), then distributed to inverters.	0.0 ... 48 000 kW (0 ... Pn_MW × 1000)
Cmd_Q	REAL	Reactive power setpoint in kVAr. Active when Cmd_VArMode=1 and VArControl_Mode=0 (fixed-Q mode). Positive = capacitive (leading); negative = inductive (lagging).	-20 000 ... +20 000 kVAr (plant P-Q envelope)
Cmd_PF	REAL	Power factor setpoint. Active when Cmd_VArMode=2. Written uniformly to all available inverters (FIX4 format: ×10000 internally). No ramp — PF is dimensionless.	-1.00 ... +1.00 Positive = lagging Negative = leading
Cmd_VArMode	INT	Reactive dispatch mode for FC_PPC_ReactiveControl. Controls <i>how</i> Q is distributed to inverters. Independent of VArControl_Mode in DB39 which controls <i>how</i> the plant-level Q command is computed.	0 = Off (zero VAr) 1 = Q dispatch (by VArAval share) 2 = PF setpoint (uniform)
LocalMode_Req	BOOL	HMI LOCAL/REMOTE selector. When TRUE, Plant_Mode = LOCAL (0); Cmd_P and Cmd_Q come from OB30 caller (HMI DB), not SCADA. Watchdog is still checked.	FALSE = REMOTE TRUE = LOCAL
f_meas	REAL	Grid frequency from analog input (4–20 mA → 47.5–52.0 Hz scaling). Active when Frequency_Source_Sel=FALSE. Used by FreqResponse for droop calculation and OFRT/UFRT trip detection.	47.5 ... 52.0 Hz Nominal: 50.0 Hz

PARAMETER	TYPE	DESCRIPTION	VALID RANGE / VALUES
Freq_Test_Override	REAL	ISCE test bench frequency input (second 4–20 mA channel, 47.5–52.0 Hz). Active when Frequency_Source_Sel=TRUE. Allows scripted frequency injection without grid events during the ANRE commissioning campaign.	47.5 ... 52.0 Hz
Frequency_Source_Sel	BOOL	Selects the active frequency source. FALSE = f_meas (real grid AI, normal operation). TRUE = Freq_Test_Override (ISCE test bench). SCADA-writable so the commissioning team can switch without a code reload.	FALSE = f_meas TRUE = test bench
U_meas	REAL	MV busbar measured voltage in kV, from grid meter or VT. Used by FB_PPC_QCapability when VArControl_Mode=1 (U-droop) to compute the reactive error: $dU = (U_{\text{setpoint}} - U_{\text{meas}}) / U_{\text{meas}}$.	0.0 ... 25.0 kV Nominal: 20.0 kV
P_actual_kW	REAL	Plant actual active power from grid meter, in kW (measurement, not setpoint). Used by FB_PPC_QCapability to index into the P-Q capability table. In Sim_Mode, set this to match Cmd_P to get a realistic operating point on the P-Q curve.	0.0 ... 48 000 kW
Sim_Mode	BOOL	Simulation mode. Forces all 10 inverters to healthy state (Enabled, RemReady, no CommError/Error/PwrOffReas/DrtStt, WAval=100%, VArAval=100%) and all skid ELECTRIC_OK=TRUE before InverterMonitor runs. Allows full PPC logic test without hardware. EN_PPC=FALSE and FALLBACK still stop the plant normally.	FALSE = normal TRUE = simulation

VAR_OUTPUT 4 parameters			
PARAMETER	TYPE	DESCRIPTION	EXPECTED IN NORMAL OPERATION
PPC_Active	BOOL	TRUE when Plant_Mode = 1 (REMOTE_IEC). Indicates the controller is actively producing setpoints from SCADA commands. Use as HMI "PPC Running" indicator.	TRUE when EN_PPC=TRUE, watchdog toggling, LocalMode_Req=FALSE
PPC_Fault	BOOL	TRUE when any inverter has CommError, active Error, or non-zero PwrOffReas. In Sim_Mode this should stay FALSE since all faults are suppressed.	FALSE in Sim_Mode TRUE if a real comms fault exists
AnyDerating	BOOL	TRUE when any healthy inverter reports a non-zero DrtStt (thermal, grid frequency, or other derating). Informational only — power is still distributed; WAval already reflects reduced capacity.	FALSE in Sim_Mode (DrtStt forced 0)
Mode_Out	INT	Current Plant_Mode for HMI display. Mirrors the internal Plant_Mode state variable.	0 = LOCAL 1 = REMOTE_IEC (normal) 2 = FALLBACK (watchdog expired)

"PPC_CONTROLLER" DB — KEY PARAMETERS FOR TESTING

These govern behavior but are not wired as inputs – set via SCADA or Watch table

DB FIELD	TYPE	PURPOSE	DEFAULT	TEST VALUES
P_RampUp P_RampDown	REAL	Active power ramp rates in kW/s. Limits how fast Cmd_P changes reach the inverters. RampDown also governs FALLBACK ramp-to-zero speed.	e.g. 100 kW/s	100 , 500 , 1000
Droop_pct	REAL	P-f droop setting in %. ISCE Tests 1–4 require tests at 8% and 10%.	8.0	8.0 , 10.0
DeadBand_mHz	REAL	Frequency dead band in mHz. dP_droop = 0 when $ df \leq \text{DeadBand}/1000$. Set to 0 for Art.117 fine-response sub-test.	200.0	200.0 , 0.0
OFRT_Trip_Hz UFRT_Trip_Hz	REAL	Over/under-frequency trip thresholds. When f_active crosses these, Trip_FreqFault=TRUE and all inverter setpoints are zeroed.	51.5 / 47.5	Keep defaults for ISCE
Reconnect_Enable	BOOL	Rising edge clears the frequency trip latch after f returns to normal band. Operator or SCADA button.	FALSE	Pulse FALSE→TRUE after trip
VArControl_Mode	INT	Q computation source inside QCapability. 0 = use Cmd_Q directly (fixed-Q). 1 = compute Q from	0	0 fixed, 1 U-droop

DB FIELD	TYPE	PURPOSE	DEFAULT	TEST VALUES
		U_meas error (U-droop). 2 = disabled (Q_final = 0).		
Q_Ramp_Rate_fast Q_Ramp_Rate_slow	REAL	Q ramp rates in kVAr/s. Selected via Q_Ramp_Fast_Sel. ISCE Test 6 requires verification of both rates.	—	Per ISCE Test 6 values
WRtg_kW	REAL	Rated active power per inverter in kW. In Sim_Mode, InverterMonitor uses WAval=10000 (100% pu) × WRtg_kW to compute PmaxPlant. Set this to the SMA datasheet value before testing.	SMA SC rated kW	4800.0 (48 MW / 10)

TEST SCENARIOS

Recommended input combinations to verify each control function

T-01

Basic enable & P setpoint ramp

EN_PPC	TRUE
Sim_Mode	TRUE
IEC_Watchdog	toggling
CycleTime_s	0.1
Cmd_P	0 → 24000 → 48000
f_meas	50.0
P_RampUp (DB)	100 kW/s

Expect: Mode_Out=1, PPC_Active=TRUE, PPC_Fault=FALSE. WSpt on each inverter ramps at 10 kW/s per unit (100/10). At Cmd_P=48000 all 10 inverters at WRtg_kW.

T-02

Watchdog timeout → FALLBACK

EN_PPCTRUE

Sim_ModeTRUE

IEC_Watchdoghold FALSE >30 s

Cmd_P48000

Expect: After 30 s Mode_Out→2 (FALLBACK), WSpt ramps to 0, OperMode→303 on all inverters. Restore watchdog toggle → Mode_Out→1, ramps restart from 0.

T-03 P-f droop — over-frequency

Frequency_Source_SelTRUE

Freq_Test_Override50.0 → 50.5

Cmd_P48000

Droop_pct (DB)8.0

DeadBand_mHz (DB)200.0

Expect: $df=+0.5\text{ Hz} \rightarrow dP_{\text{droop}} = -(2 \times 48000 \times 0.5)/8 = -6000\text{ kW}$. $P_{\text{final}} = 42000\text{ kW}$. dP_{droop} visible in DB. No ramp delay — immediate.

T-04 Frequency trip & reconnect

Frequency_Source_SelTRUE

Freq_Test_Override51.6 (above 51.5)

Cmd_P48000

Expect: Trip_FreqFault=TRUE, WSpt→0 immediately, Ramps_Pcmd reset to 0. Return Freq_Test_Override to 50.0, then pulse Reconnect_Enable TRUE→FALSE. Trip_FreqFault clears, Reconnecting=TRUE, P ramps from 0. Reconnect_Timer_s measures recovery time (ISCE Test 7).

T-05 Fixed-Q reactive dispatch

Cmd_VArMode1

VArControl_Mode (DB)0 (fixed Q)

Cmd_Q10000 kVAr

P_actual_kW24000 (50% Pn)

Expect: Q ramps to 10000 kVAr at Q_Ramp_Rate. VArSpt distributed proportionally across 10 inverters (1000 kVAr each if equal VArAval). Q_limited=FALSE if within P-Q envelope at 50% P.

T-06 U-droop voltage control

Cmd_VArMode1

VARControl_Mode (DB)	1 (U droop)
U_setpoint_ext (DB)	20.5 kV
U_meas	20.0 kV
U_Droop_pct (DB)	5.0

Expect: $dU_{pu} = 0.5/20 = 0.025$. $Q_{command} = (0.025/0.05) \times Q_{maxPlant} = 50\% Q_{max}$ (capacitive). Q ramp applies. Increase U_meas → Q command decreases.

T-07 PF setpoint mode

Cmd_VArMode	2
Cmd_PF	0.95
Cmd_P	24000

Expect: $PFSpt = REAL_TO_DINT(0.95 \times 10000) = 9500$ written to all available inverters. VArMode=1075 (PFctlCom). No Q ramp — PF is dimensionless.

T-08 Disable (EN_PPC = FALSE)

EN_PPC	TRUE → FALSE
Cmd_P	48000 (was running)

Expect: Immediate OperMode=303 (Stop) on all inverters, WSpt=0, VArSpt=0. Ramps_Pcmd reset to 0, Trip_Latch cleared. Mode_Out=0. PPC_Active=FALSE. Re-enable → ramps from 0.